

# How Much Can Data Augmentation Compensate for Leakage-Safe Splitting?

A Factorial Experiment on CWRU Bearing Fault Diagnosis

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## Abstract

Sliding-window preprocessing in bearing diagnosis creates adjacent segments that share samples. Random window splitting can place that shared content across partitions, whereas recording-level splitting prevents any source file from crossing partitions. We test whether augmentation compensates for the resulting operational performance gap using a  $2 \times 2 \times 10 \times 5$  factorial experiment (two CNNs, two split protocols, ten augmentation conditions, five seeds; 200 independent training runs) on CWRU. Without augmentation, moving from random-window to recording-level evaluation reduces accuracy by 5.69 percentage points for 1D-CNN and 11.43 points for 2D-CNN (pooled-SD Cohen’s  $d = 0.93$  and  $1.47$ ). A 5-sample time shift gives the best recording-level result for both models and recovers 52.55% and 32.63% of those protocol gaps. The destructive FFT-magnitude-reversal control collapses recording-level accuracy to 0.403 and 0.392. H3 is not supported: physical-fidelity correlations are non-significant for 1D ( $\rho = 0.433$ ,  $p = 0.244$ ) and 2D ( $\rho = 0.317$ ,  $p = 0.406$ ), and remain non-significant after removing the negative control. Fault-size grouping produces the lowest descriptive stress-test accuracies (0.651 and 0.492), but unequal group counts and training coverage preclude a causal ranking against load or recording grouping. Exact manifests, 200 epoch logs, confusion matrices, and generated tables are released for verification.

## 1 Introduction

Vibration-based deep learning for bearing fault diagnosis has achieved near-perfect accuracy on benchmark datasets [1–3]. However, fixed-length sliding windows extracted from continuous recordings are not independent: adjacent windows can share signal samples. When pooled windows are randomly assigned to train, validation, and test sets, windows from the same source recording can cross partitions. A classifier may then exploit shared samples or recording-specific signatures, so the evaluation can overstate performance on unseen recordings [4, 5].

This data leakage is a known problem in time-series evaluation [4, 5], yet its interaction with data augmentation—a standard regularization technique—has not been systematically studied. Augmentation increases training diversity and could, in principle, compensate for the reduced diversity under leakage-safe splitting. But augmentation may also distort physically diagnostic frequency bands, undermining the very features a classifier should learn.

The concrete bottleneck is therefore not benchmark accuracy itself, but separating the apparent benefit of augmentation from the evaluation protocol that measures it. Our research question is: under matched preprocessing, how much of the random-to-recording performance gap can augmentation recover, and is that recovery associated with preservation of fault-frequency-band energy?

We pose three hypotheses:

- **H1:** Data augmentation partially recovers the performance gap between leakage-prone (random window) and leakage-safe (recording-level) splits.

- **H2:** The random-to-recording protocol gap dominates the augmentation gain—even the best augmentation cannot restore random-split performance.
- **H3:** Augmentations that better preserve fault-frequency-band energy yield greater recovery (physical fidelity hypothesis).

To test these hypotheses, we conduct a fully factorial experiment on the Case Western Reserve University (CWRU) bearing dataset: 2 model architectures (1D-CNN on raw waveform, 2D-CNN on STFT spectrograms)  $\times$  2 split protocols  $\times$  10 augmentation conditions  $\times$  5 random seeds, totaling 200 independent training runs. We also run descriptive stress tests based on recording identity, motor load, and fault size; their unequal group counts are reported explicitly.

The contribution is threefold: (1) a matched-preprocessing estimate of the random-window versus recording-level protocol gap; (2) a gap-recovery analysis across two model families, nine augmentations, and a destructive control; and (3) a traceable evidence package that connects physical-fidelity audits, per-class metrics, exact split assignments, and complete epoch logs to the manuscript. These are benchmark-scoped methodological contributions, not claims of industrial generalization.

The dataset comprises 40 recording files yielding 11,832 windows (1,024 samples per window, 50% overlap). Each configuration trains for 50 epochs on an NVIDIA A10 GPU (Adam optimizer,  $\text{lr} = 0.001$ , batch size 128).

## 2 Related Work

**CWRU benchmark.** The Case Western Reserve University bearing dataset [6] is a widely used benchmark in vibration-based fault diagnosis. Smith and Randall [1] analyzed diagnostic use of the CWRU data and documented benchmark-specific caveats. Wen et al. [2] and Zhang et al. [3] illustrate CNN-based diagnosis from vibration signals; neither citation by itself validates the split protocol used in the present study.

**Augmentation inspiration.** SpecAugment [7], originally proposed for speech recognition, applies time and frequency masking to spectrograms. We cite it as methodological inspiration only: our condition masks real-FFT bins and contiguous waveform samples before model-specific representation, and is therefore a waveform-compatible adaptation rather than a reproduction of the speech method.

**Dependence-aware evaluation.** Bergmeir and Benítez [4] analyze when cross-validation is appropriate for time-series predictors, while Cerqueira et al. [5] compare performance-estimation procedures for forecasting. These studies do not directly settle bearing classification, but they motivate an explicit dependence audit. Here the risk is concrete: overlapping windows from one source recording can cross partitions under window-level randomization.

**Cross-condition diagnosis and physical constraints.** Transfer-learning studies such as Shao et al. [8] and Li et al. [9] motivate evaluation beyond a single observed condition. Separately, vibration frequency axes carry physical units, so arbitrary image-style transforms require validation. We therefore measure fault-band energy fidelity directly and do not use the transfer-learning citations as evidence that a particular augmentation is physically safe.

## 3 Dataset and Leakage Control

### 3.1 CWRU Bearing Dataset

We use the CWRU 12 kHz Drive End bearing dataset, comprising 40 .mat recording files across four fault classes: Normal (4 recordings), Inner Race fault (12), Outer Race fault @6 o'clock (12), and Ball fault (12). Fault diameters include 0.007, 0.014, and 0.021 inches. Motor loads range from 0 to 3 HP.

The files were downloaded on 17 July 2026 from the public CWRU Bearing Data Center page [6]. The source page does not state a formal license in the locally retained documentation, so this release records the access terms and file hashes but does not redistribute raw signals. The exact selected filenames, byte sizes, SHA-256 values, and channel metadata are stored in `data_manifest/dataset_file_manifest.json`.

Each recording is segmented into fixed-length windows of 1,024 points with 50% overlap, yielding 11,832 windows total. The fault-class distribution is: Normal 952, IR 3,624, OR 3,582, Ball 3,674.

### 3.2 Leakage Audit

**Leakage source:** Sliding-window overlap. Adjacent windows from the same recording share up to 512 points (50% overlap). Random splitting distributes these correlated windows across train/val/test, enabling a classifier to recognize recording identity via temporal proximity.

**Leakage control:** Recording-level split. All windows from a given .mat file are assigned entirely to train, validation, or test, stratified by fault class. Train/val/test ratio is 60/20/20. No recording appears in more than one split.

We verify the leakage path through a structural adjacent-window audit (Section 5, Figure 1). With the fixed 50% overlap used in both protocols, adjacent windows share exactly 512 samples; aligned shared regions are identical. Random splitting can place these duplicated samples across partitions, whereas recording-level splitting cannot.

**Safe practices:** Window length and 50% overlap are identical across protocols, so only the split unit changes. Z-score normalization statistics are computed on the training set only and applied to validation/test data. For each augmented run, the seeded transform is sampled once after splitting and replaces the normalized training windows; it is not resampled per epoch or concatenated with the originals. Validation and test data remain unaugmented. Generated manifests record all selected-file hashes and all five recording-level assignments.

## 4 Method

### 4.1 Experimental Design

We employ a fully crossed factorial design:

- **Factor A—Model:** 1D-CNN (4 conv layers, raw waveform input  $1 \times 1024$ ) and 2D-CNN (4 conv layers, STFT spectrogram input  $1 \times 32 \times 32$ ,  $n_{\text{fft}} = 128$ ,  $\text{hop} = 64$ )
- **Factor B—Split Protocol:** Random window split (leakage baseline) and Recording-level split (leakage-safe)
- **Factor C—Augmentation:** 10 conditions: none; Gaussian noise ( $\sigma = 0.01, 0.05$ , or  $0.10$ ); time shift (5, 20, or 50 samples); waveform-compatible spectral/time masking; combined noise ( $\sigma = 0.05$ ) plus shift 20; and FFT-magnitude frequency-axis reversal (negative control)
- **Factor D—Random Seed:** 5 seeds (42, 123, 456, 789, 1024)

### 4.2 Grouping-Variable Ablation

We additionally compare three grouping strategies for train/val/test splitting (no augmentation, 5 seeds each):

- **By Recording:** groups defined by .mat file identity, yielding 23/7/10 train/validation/test recordings
- **By Load:** four motor-load groups (0, 1, 2, 3 HP), assigned as two/one/one groups and yielding 20/10/10 recordings

- **By Fault Size:** three fault-diameter groups (0.007, 0.014, 0.021 inches), assigned as one/one/one groups; the four Normal recordings are assigned separately as two/one/one, yielding 14/13/13 recordings

All strategies use the same 50%-overlap segmentation and training code. However, the number of groups and the resulting training coverage are unequal across strategies. We therefore treat their accuracies as descriptive stress-test results, not as a causal ranking of domain-shift variables.

### 4.3 Scope Relative to the Initial Plan

The initial research brief proposed a fully crossed contamination study and all 12 train-load/test-load pairs. The audited release limits primary inference to the 200-run model  $\times$  split  $\times$  augmentation factorial design. The 30 grouping runs are descriptive because their training coverage is unequal, and the paired contamination check uses one common model seed and common noise realizations only. It is retained as an exploratory artifact, not as a robustness-superiority test. The mechanism definitions were also corrected before the audited rerun: M2 uses deterministic input-representation distance rather than an untrained hidden layer, and M5 uses symmetric energy fidelity rather than an unbounded energy ratio.

## 5 Mechanism Validation

Following the evidence hierarchy outlined in the course manual, we conduct three mechanism-validation experiments *before* classification evaluation:

### 5.1 Structural Adjacent-Window Audit (M1)

For offsets 1–10, we compute the exact shared-sample fraction, correlation and mean absolute error after aligning the shared region, and zero-lag Pearson similarity. At offset one, the 512 shared samples are identical by construction (50% of each window); zero-lag similarity remains data dependent. Figure 1 separates guaranteed sample duplication from ordinary waveform correlation and establishes a leakage path without claiming that Pearson correlation alone proves model exploitation.

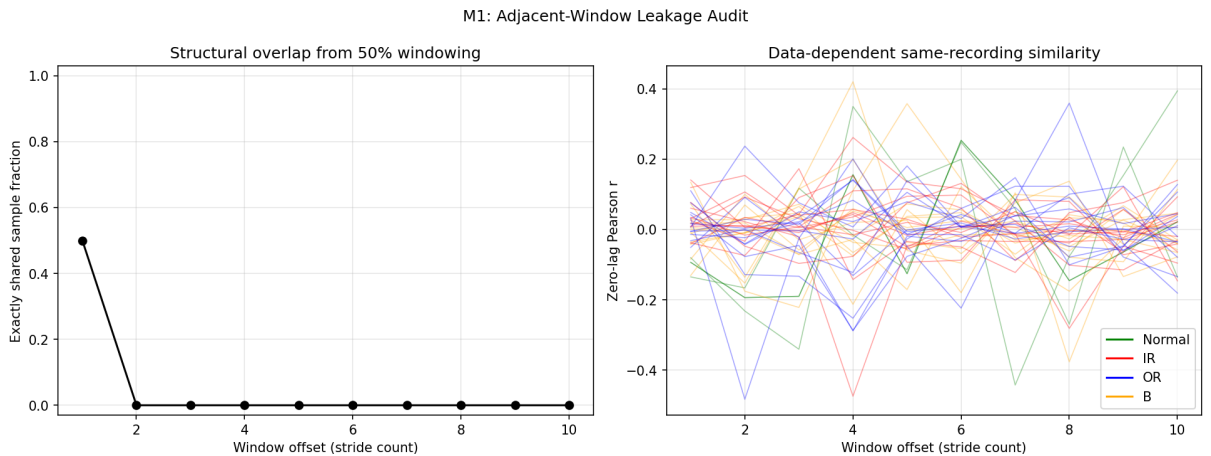


Figure 1: M1: Structural adjacent-window overlap and data-dependent zero-lag similarity.

## 5.2 Fault-Band Energy Audit (M5)

For each augmentation, we compute a symmetric energy-fidelity score

$$F_{\text{energy}} = \min \left( \frac{E_{\text{aug}}}{E_{\text{orig}}}, \frac{E_{\text{orig}}}{E_{\text{aug}}} \right)$$

in envelope-spectrum neighborhoods around the first three harmonics of the relevant bearing frequency (BPFO, BPFI, or BSF). The score lies in  $[0, 1]$  and penalizes attenuation and amplification equally, avoiding the error of treating broadband energy injection as preservation.

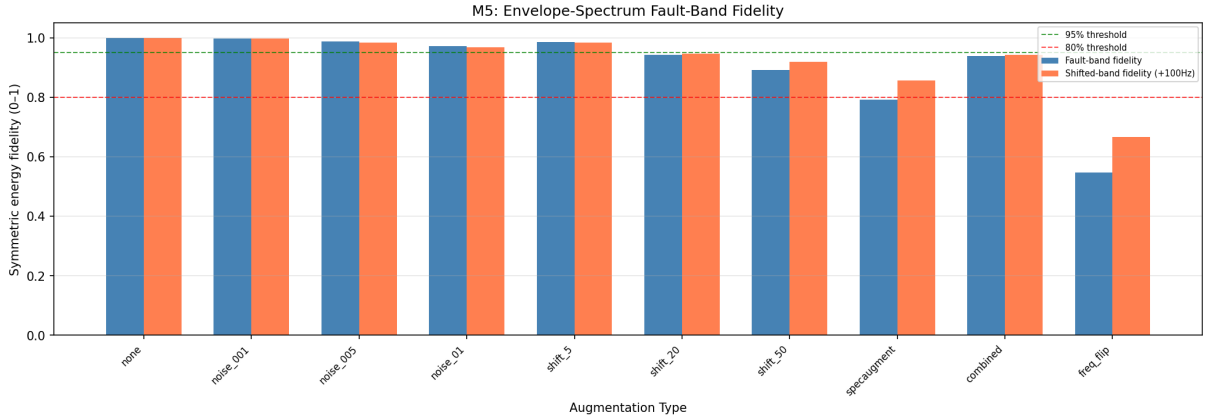


Figure 2: M5: Symmetric envelope-spectrum fault-band energy fidelity ( $F_{\text{energy}}$ ).

## 5.3 Feature Diversity Analysis (M2)

For 200 paired training samples, we compute standardized  $32 \times 32$  STFT magnitude representations before and after augmentation, then measure paired cosine distance and SVD effective rank (Figure 3). This deterministic input-representation audit avoids attributing meaning to an untrained network’s hidden layer.

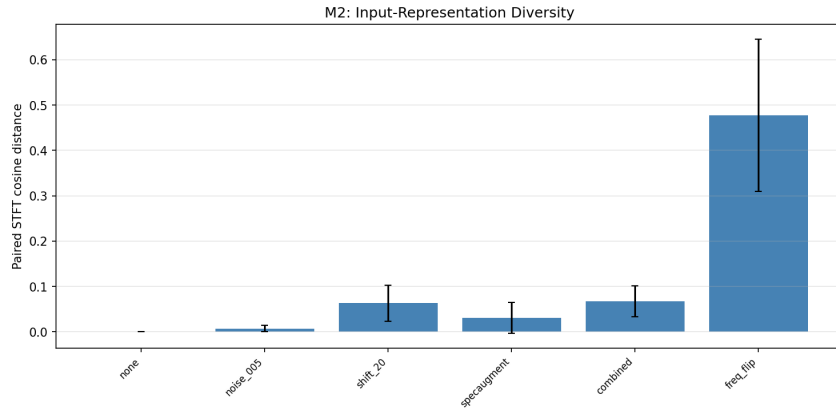


Figure 3: M2: Input-representation diversity (paired STFT cosine distance).

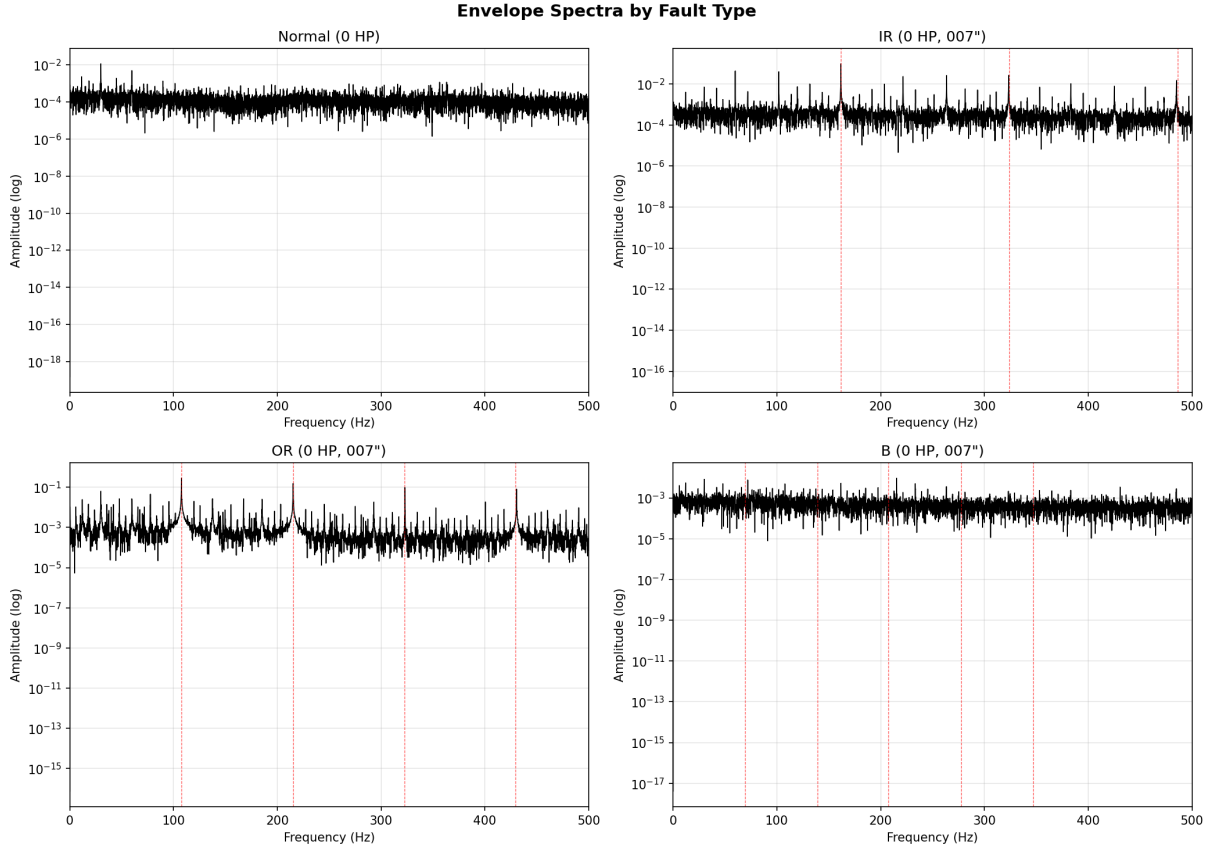


Figure 4: Envelope spectrum examples for each fault class.

## 6 Experimental Setup

- **Hardware:** Baidu Cloud BCC, NVIDIA A10 (24 GB), Ubuntu 22.04; PyTorch 2.5.1+cu121
- **Training:** Adam ( $\text{lr} = 0.001$ ), CrossEntropy loss, `batch_size=128`, 50 epochs
- **Normalization:** Z-score computed on training set only
- **Metrics:** Accuracy, macro-F1, per-class precision/recall/F1
- **Statistics:** Mean  $\pm$  std across 5 seeds; Cohen's  $d$  for effect size
- **Best-validation epoch diagnostic:** mean 12.58, median 7, maximum 50; 8/200 runs attain their best validation accuracy in epochs 45–50 (Figure 5).

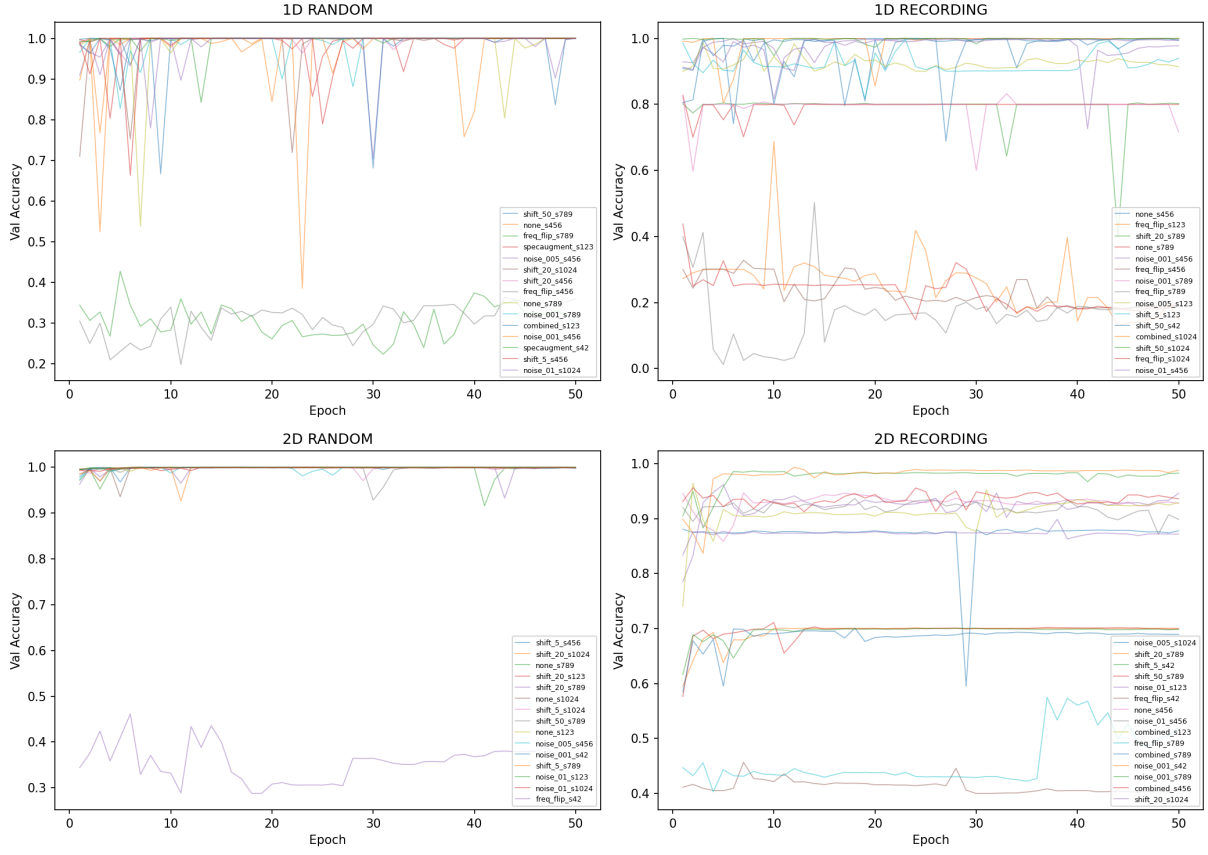


Figure 5: Convergence curves for all 200 training runs across four configurations.

## 7 Results

### 7.1 Main Factorial Results

Table 1 reports all 40 configurations. The no-augmentation random-window baselines are 0.9998 (1D) and 0.9999 (2D), compared with 0.9429 and 0.8856 under recording-level splitting. The corresponding protocol gaps are 0.0569 and 0.1143; pooled-SD Cohen’s  $d$  is 0.9262 and 1.4697 (Figure 6). The physically plausible random-split augmentations remain near ceiling, whereas the destructive negative control drops to 0.4069 and 0.4488.

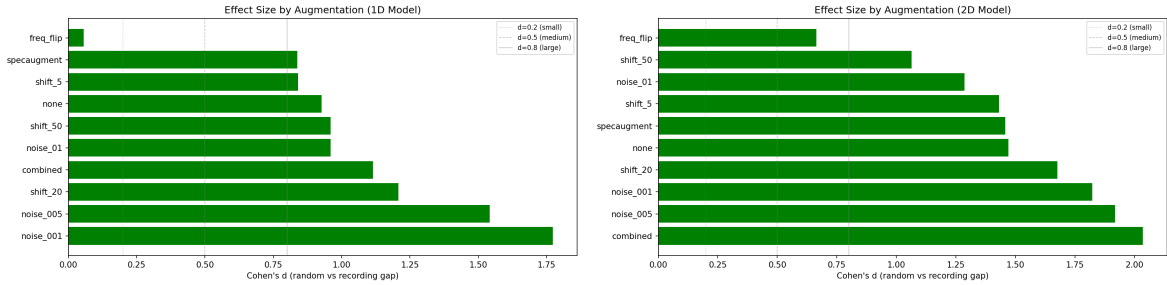


Figure 6: Cohen’s  $d$  effect sizes. Left: 1D-CNN. Right: 2D-CNN.

Table 2 reports the required Macro-F1 results for all 40 configurations. Under recording-level splitting, Macro-F1 changes from 0.9365 to 0.9699 for 1D and from 0.8717 to 0.9187 for 2D when the 5-sample shift is applied; the destructive control falls to 0.3337 and 0.2911.

For 1D-CNN under recording-level splitting, three of nine augmentations improve on the

Table 1: Main results: accuracy (mean  $\pm$  std) across 5 seeds.

| Augmentation           | 1D Random           | 1D Recording        | 2D Random           | 2D Recording        |
|------------------------|---------------------|---------------------|---------------------|---------------------|
| None                   | 0.9998 $\pm$ 0.0002 | 0.9429 $\pm$ 0.0777 | 0.9999 $\pm$ 0.0002 | 0.8856 $\pm$ 0.0984 |
| Noise 0.01             | 0.9999 $\pm$ 0.0002 | 0.9379 $\pm$ 0.0443 | 0.9997 $\pm$ 0.0003 | 0.8807 $\pm$ 0.0826 |
| Noise 0.05             | 0.9999 $\pm$ 0.0002 | 0.9628 $\pm$ 0.0305 | 0.9998 $\pm$ 0.0002 | 0.8900 $\pm$ 0.0724 |
| Noise 0.10             | 0.9997 $\pm$ 0.0005 | 0.9351 $\pm$ 0.0851 | 0.9997 $\pm$ 0.0003 | 0.9188 $\pm$ 0.0795 |
| Shift 5                | 0.9999 $\pm$ 0.0002 | 0.9728 $\pm$ 0.0407 | 0.9997 $\pm$ 0.0005 | 0.9229 $\pm$ 0.0679 |
| Shift 20               | 0.9997 $\pm$ 0.0004 | 0.9267 $\pm$ 0.0764 | 0.9997 $\pm$ 0.0003 | 0.9091 $\pm$ 0.0683 |
| Shift 50               | 0.9999 $\pm$ 0.0002 | 0.9396 $\pm$ 0.0795 | 0.9997 $\pm$ 0.0005 | 0.8477 $\pm$ 0.1808 |
| Spectral/time mask     | 0.9999 $\pm$ 0.0002 | 0.9439 $\pm$ 0.0845 | 0.9995 $\pm$ 0.0005 | 0.9009 $\pm$ 0.0856 |
| Combined               | 0.9999 $\pm$ 0.0002 | 0.9334 $\pm$ 0.0755 | 0.9997 $\pm$ 0.0005 | 0.8960 $\pm$ 0.0645 |
| FFT magnitude reversal | 0.4069 $\pm$ 0.0377 | 0.4026 $\pm$ 0.0882 | 0.4488 $\pm$ 0.0369 | 0.3919 $\pm$ 0.1019 |

Table 2: Macro-F1 (mean  $\pm$  std) across 5 seeds for all factorial configurations.

| Augmentation           | 1D Random           | 1D Recording        | 2D Random           | 2D Recording        |
|------------------------|---------------------|---------------------|---------------------|---------------------|
| None                   | 0.9998 $\pm$ 0.0002 | 0.9365 $\pm$ 0.0872 | 0.9999 $\pm$ 0.0002 | 0.8717 $\pm$ 0.1146 |
| Noise 0.01             | 0.9999 $\pm$ 0.0002 | 0.9393 $\pm$ 0.0410 | 0.9997 $\pm$ 0.0003 | 0.8771 $\pm$ 0.0804 |
| Noise 0.05             | 0.9999 $\pm$ 0.0002 | 0.9618 $\pm$ 0.0342 | 0.9998 $\pm$ 0.0002 | 0.8868 $\pm$ 0.0724 |
| Noise 0.10             | 0.9997 $\pm$ 0.0005 | 0.9173 $\pm$ 0.1217 | 0.9997 $\pm$ 0.0003 | 0.9145 $\pm$ 0.0801 |
| Shift 5                | 0.9999 $\pm$ 0.0002 | 0.9699 $\pm$ 0.0473 | 0.9997 $\pm$ 0.0005 | 0.9187 $\pm$ 0.0700 |
| Shift 20               | 0.9996 $\pm$ 0.0004 | 0.9159 $\pm$ 0.0964 | 0.9997 $\pm$ 0.0003 | 0.9055 $\pm$ 0.0689 |
| Shift 50               | 0.9999 $\pm$ 0.0002 | 0.9430 $\pm$ 0.0712 | 0.9997 $\pm$ 0.0005 | 0.8337 $\pm$ 0.2010 |
| Spectral/time mask     | 0.9999 $\pm$ 0.0002 | 0.9247 $\pm$ 0.1239 | 0.9995 $\pm$ 0.0005 | 0.8946 $\pm$ 0.0889 |
| Combined               | 0.9999 $\pm$ 0.0002 | 0.9193 $\pm$ 0.1059 | 0.9997 $\pm$ 0.0005 | 0.8951 $\pm$ 0.0599 |
| FFT magnitude reversal | 0.3527 $\pm$ 0.0526 | 0.3337 $\pm$ 0.1076 | 0.3414 $\pm$ 0.0460 | 0.2911 $\pm$ 0.0977 |



0.9429 baseline; for 2D-CNN, six improve on the 0.8856 baseline. A 5-sample shift is best for both models (0.9728 and 0.9229). The frequency-reversal negative control performs worst (0.4026 and 0.3919), supporting its role as a destructive check. Figures 7–8 visualize the contrast.

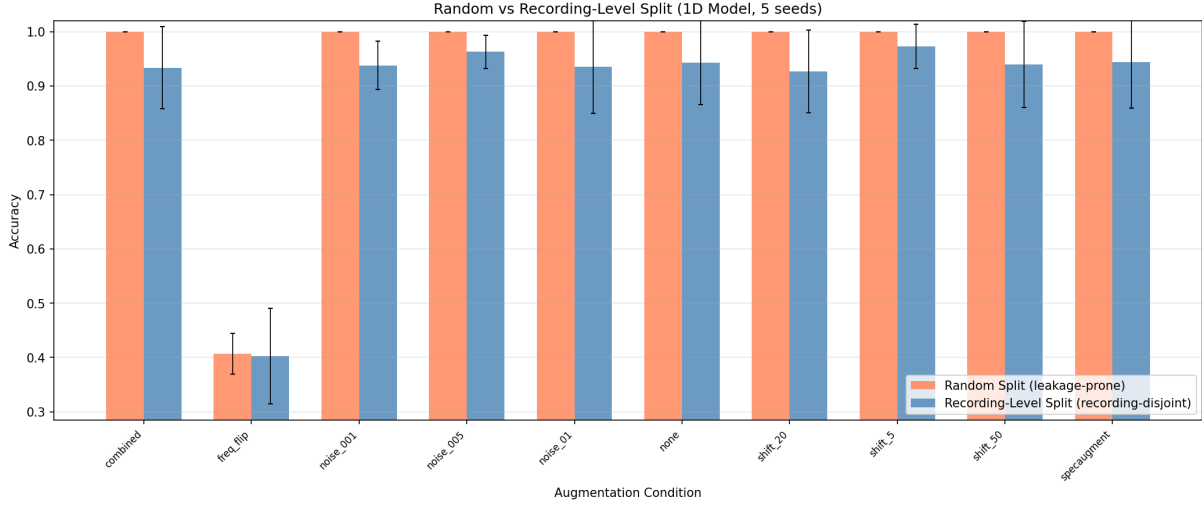


Figure 7: Random vs. recording-level split accuracy (1D-CNN).

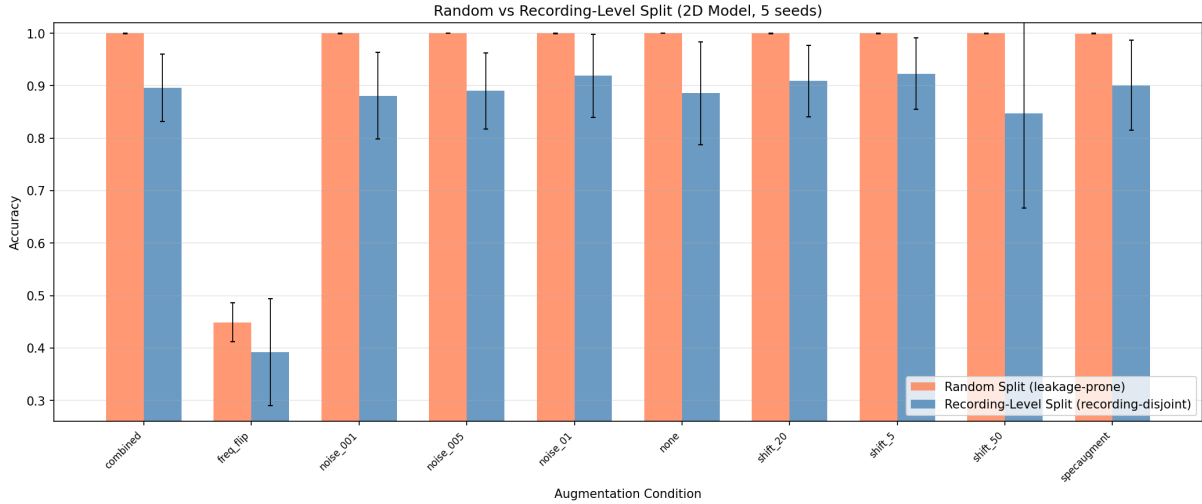


Figure 8: Random vs. recording-level split accuracy (2D-CNN).

## 7.2 Per-Class Analysis

Table 3 reports 2D-CNN recording-level per-class F1. Excluding the deliberately destructive negative control, Normal is 0.9996–1.0000, while IR, OR, and Ball span 0.742–0.861, 0.730–0.892, and 0.753–0.943. Under frequency reversal, IR F1 falls to zero and the other classes also deteriorate sharply.

Table 4 adds per-class precision and recall for both models under the no-augmentation baseline, the best augmentation, and the destructive control. This prevents the aggregate metrics from hiding asymmetric class failures; in particular, the negative control drives 2D-CNN IR recall to zero.

Per-class heatmaps for both models are provided in Figure 9.

Table 3: Per-class F1 under recording-level splitting for the 2D-CNN (mean  $\pm$  std).

| Augmentation           | Normal            | IR                | OR                | Ball              |
|------------------------|-------------------|-------------------|-------------------|-------------------|
| None                   | 1.000 $\pm$ 0.000 | 0.742 $\pm$ 0.293 | 0.838 $\pm$ 0.137 | 0.906 $\pm$ 0.057 |
| Noise 0.01             | 1.000 $\pm$ 0.000 | 0.817 $\pm$ 0.199 | 0.827 $\pm$ 0.111 | 0.864 $\pm$ 0.091 |
| Noise 0.05             | 1.000 $\pm$ 0.000 | 0.803 $\pm$ 0.192 | 0.858 $\pm$ 0.116 | 0.886 $\pm$ 0.076 |
| Noise 0.10             | 1.000 $\pm$ 0.000 | 0.843 $\pm$ 0.202 | 0.891 $\pm$ 0.117 | 0.924 $\pm$ 0.061 |
| Shift 5                | 1.000 $\pm$ 0.001 | 0.848 $\pm$ 0.184 | 0.884 $\pm$ 0.104 | 0.943 $\pm$ 0.069 |
| Shift 20               | 1.000 $\pm$ 0.000 | 0.831 $\pm$ 0.190 | 0.877 $\pm$ 0.099 | 0.914 $\pm$ 0.053 |
| Shift 50               | 1.000 $\pm$ 0.000 | 0.851 $\pm$ 0.172 | 0.730 $\pm$ 0.371 | 0.753 $\pm$ 0.286 |
| Spectral/time mask     | 1.000 $\pm$ 0.000 | 0.802 $\pm$ 0.230 | 0.859 $\pm$ 0.112 | 0.918 $\pm$ 0.052 |
| Combined               | 1.000 $\pm$ 0.000 | 0.861 $\pm$ 0.109 | 0.850 $\pm$ 0.118 | 0.870 $\pm$ 0.079 |
| FFT magnitude reversal | 0.568 $\pm$ 0.205 | 0.000 $\pm$ 0.000 | 0.453 $\pm$ 0.272 | 0.144 $\pm$ 0.132 |

Table 4: Recording-level per-class precision, recall, and F1 (mean  $\pm$  std across 5 seeds) for the baseline, best augmentation, and destructive control.

| Model  | Augmentation           | Metric    | Normal            | IR                | OR                | Ball              |
|--------|------------------------|-----------|-------------------|-------------------|-------------------|-------------------|
| CNN-1D | None                   | Precision | 0.997 $\pm$ 0.007 | 0.947 $\pm$ 0.096 | 0.934 $\pm$ 0.116 | 0.958 $\pm$ 0.085 |
|        |                        | Recall    | 1.000 $\pm$ 0.000 | 0.939 $\pm$ 0.121 | 1.000 $\pm$ 0.000 | 0.841 $\pm$ 0.294 |
|        |                        | F1        | 0.998 $\pm$ 0.003 | 0.934 $\pm$ 0.077 | 0.962 $\pm$ 0.068 | 0.852 $\pm$ 0.229 |
|        | Shift 5                | Precision | 0.994 $\pm$ 0.012 | 0.987 $\pm$ 0.016 | 0.998 $\pm$ 0.004 | 0.937 $\pm$ 0.125 |
|        |                        | Recall    | 1.000 $\pm$ 0.000 | 0.906 $\pm$ 0.187 | 1.000 $\pm$ 0.000 | 0.978 $\pm$ 0.022 |
|        |                        | F1        | 0.997 $\pm$ 0.006 | 0.932 $\pm$ 0.119 | 0.999 $\pm$ 0.002 | 0.952 $\pm$ 0.070 |
|        | FFT magnitude reversal | Precision | 0.522 $\pm$ 0.443 | 0.226 $\pm$ 0.169 | 0.275 $\pm$ 0.075 | 0.788 $\pm$ 0.395 |
|        |                        | Recall    | 0.600 $\pm$ 0.490 | 0.336 $\pm$ 0.322 | 0.512 $\pm$ 0.319 | 0.124 $\pm$ 0.117 |
|        |                        | F1        | 0.554 $\pm$ 0.458 | 0.253 $\pm$ 0.214 | 0.325 $\pm$ 0.144 | 0.202 $\pm$ 0.183 |
| CNN-2D | None                   | Precision | 1.000 $\pm$ 0.000 | 0.940 $\pm$ 0.078 | 0.822 $\pm$ 0.196 | 0.868 $\pm$ 0.111 |
|        |                        | Recall    | 1.000 $\pm$ 0.000 | 0.696 $\pm$ 0.360 | 0.884 $\pm$ 0.113 | 0.960 $\pm$ 0.041 |
|        |                        | F1        | 1.000 $\pm$ 0.000 | 0.742 $\pm$ 0.293 | 0.838 $\pm$ 0.137 | 0.906 $\pm$ 0.057 |
|        | Shift 5                | Precision | 1.000 $\pm$ 0.000 | 0.969 $\pm$ 0.031 | 0.879 $\pm$ 0.163 | 0.925 $\pm$ 0.101 |
|        |                        | Recall    | 0.999 $\pm$ 0.002 | 0.803 $\pm$ 0.260 | 0.926 $\pm$ 0.130 | 0.965 $\pm$ 0.035 |
|        |                        | F1        | 1.000 $\pm$ 0.001 | 0.848 $\pm$ 0.184 | 0.884 $\pm$ 0.104 | 0.943 $\pm$ 0.069 |
|        | FFT magnitude reversal | Precision | 0.438 $\pm$ 0.168 | 0.000 $\pm$ 0.000 | 0.367 $\pm$ 0.244 | 0.354 $\pm$ 0.135 |
|        |                        | Recall    | 0.843 $\pm$ 0.301 | 0.000 $\pm$ 0.000 | 0.624 $\pm$ 0.339 | 0.120 $\pm$ 0.119 |
|        |                        | F1        | 0.568 $\pm$ 0.205 | 0.000 $\pm$ 0.000 | 0.453 $\pm$ 0.272 | 0.144 $\pm$ 0.132 |

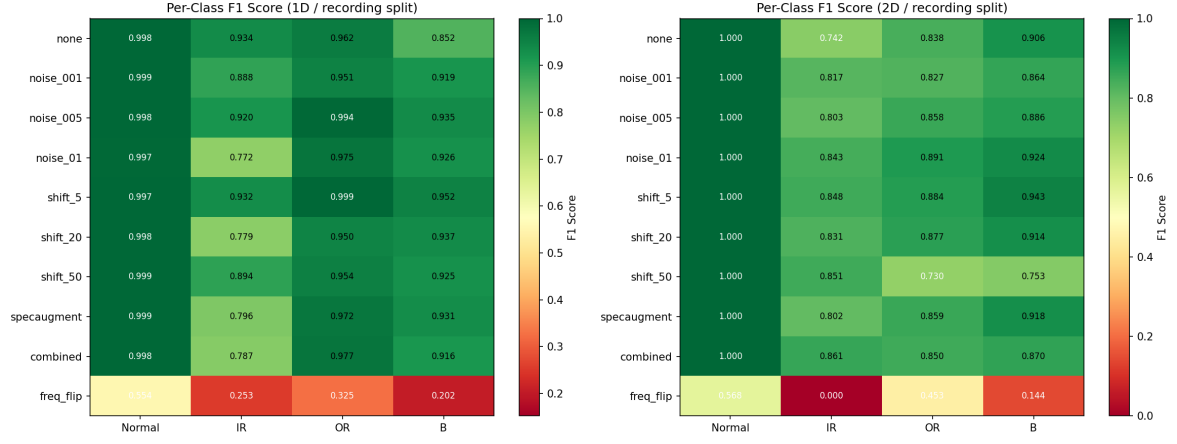


Figure 9: Per-class F1 heatmaps. Left: 1D-CNN. Right: 2D-CNN.

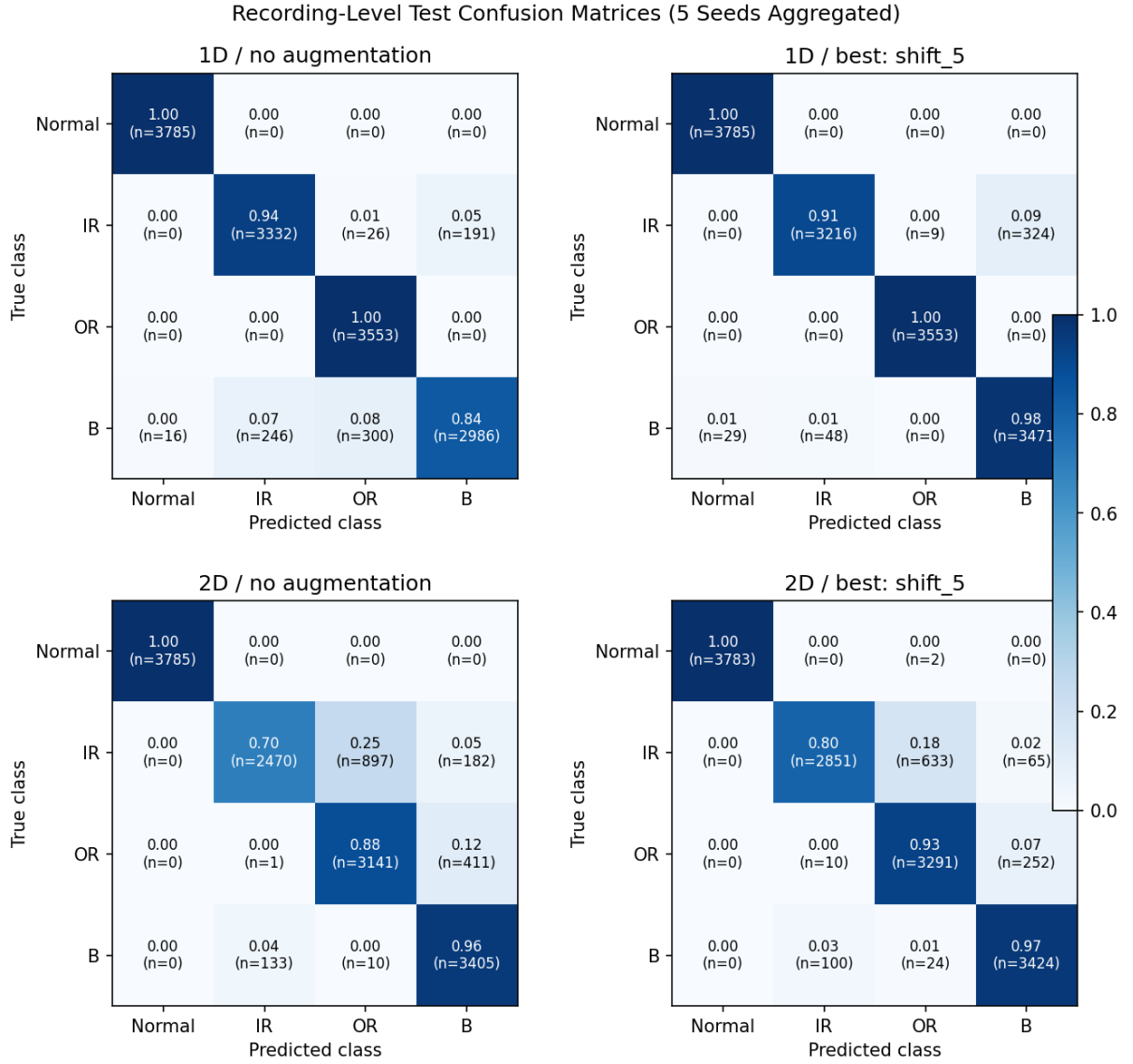


Figure 10: Row-normalized recording-level test confusion matrices aggregated across five seeds for selected baseline and augmentation conditions. Cell labels also report raw aggregated counts.

### 7.3 Gap-Recovery Analysis

Table 5 reports  $\Delta_{\text{rec}}$ —the proportion of the no-augmentation random-to-recording protocol gap recovered by each augmentation. The best values are 0.5255 (1D) and 0.3263 (2D), both from a 5-sample shift.

Table 5: Gap recovery relative to the no-augmentation random-to-recording protocol gap.

| Augmentation           | Gap (1D) | $\Delta_{\text{rec}}$ (1D) | $d$ (1D) | Gap (2D) | $\Delta_{\text{rec}}$ (2D) | $d$ (2D) |
|------------------------|----------|----------------------------|----------|----------|----------------------------|----------|
| None                   | 0.0569   | +0.0000                    | 0.9262   | 0.1143   | +0.0000                    | 1.4697   |
| Noise 0.01             | 0.0620   | -0.0879                    | 1.7729   | 0.1190   | -0.0429                    | 1.8234   |
| Noise 0.05             | 0.0371   | +0.3497                    | 1.5421   | 0.1098   | +0.0385                    | 1.9189   |
| Noise 0.10             | 0.0646   | -0.1371                    | 0.9600   | 0.0809   | +0.2905                    | 1.2869   |
| Shift 5                | 0.0271   | +0.5255                    | 0.8412   | 0.0768   | +0.3263                    | 1.4306   |
| Shift 20               | 0.0730   | -0.2847                    | 1.2076   | 0.0906   | +0.2056                    | 1.6770   |
| Shift 50               | 0.0603   | -0.0580                    | 0.9596   | 0.1520   | -0.3316                    | 1.0632   |
| Spectral/time mask     | 0.0560   | +0.0176                    | 0.8384   | 0.0986   | +0.1339                    | 1.4576   |
| Combined               | 0.0665   | -0.1670                    | 1.1148   | 0.1037   | +0.0910                    | 2.0354   |
| FFT magnitude reversal | 0.0043   | -9.4956                    | 0.0567   | 0.0569   | -4.3193                    | 0.6640   |

Figures 11–12 visualize the gap-recovery landscape.

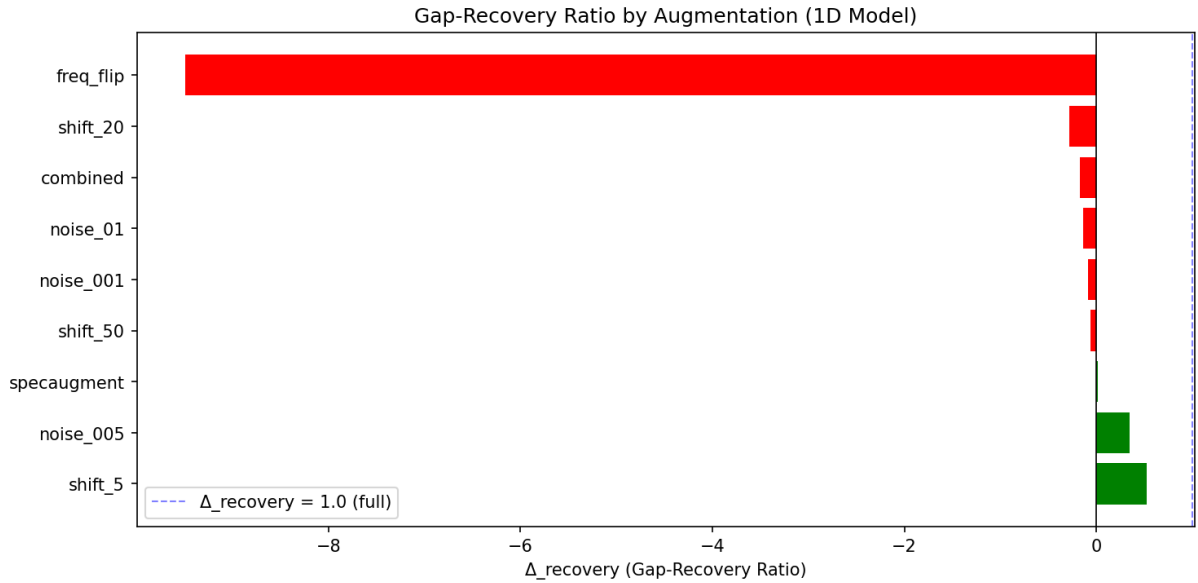


Figure 11: Gap-recovery ratio by augmentation (1D-CNN).

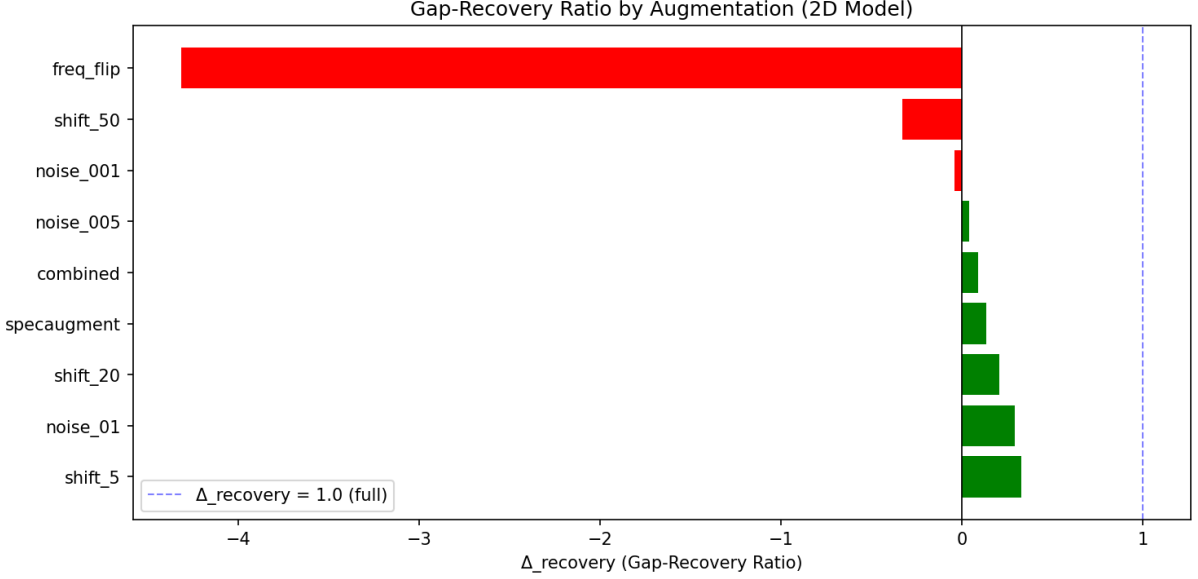


Figure 12: Gap-recovery ratio by augmentation (2D-CNN).

#### 7.4 Grouping-Variable Ablation

Table 6 compares no-augmentation accuracy across the three grouping stress tests. Fault-size grouping is lowest (0.6508 for 1D and 0.4920 for 2D), compared with recording grouping (0.9429 and 0.8856) and load grouping (0.9607 and 0.9779).

Table 6: Descriptive grouping stress-test accuracy (mean  $\pm$  std across 5 seeds). Load and fault-size strategies use unequal group counts and are not causally comparable.

| Model  | By Load (2/1/1)     | By Recording        | By Fault Size (1/1/1) |
|--------|---------------------|---------------------|-----------------------|
| CNN-1D | 0.9607 $\pm$ 0.0331 | 0.9429 $\pm$ 0.0777 | 0.6508 $\pm$ 0.1131   |
| CNN-2D | 0.9779 $\pm$ 0.0286 | 0.8856 $\pm$ 0.0984 | 0.4920 $\pm$ 0.1182   |

This ordering is descriptive, not causal. Recording, load, and fault-size strategies contain 23/7/10, 20/10/10, and 14/13/13 train/validation/test recordings, respectively; load also uses 2/1/1 groups, whereas fault size uses 1/1/1. Lower fault-size accuracy is therefore consistent with a difficult stress test but is confounded by reduced training coverage.

## 8 Efficiency / Robustness / Physical Consistency

H3 is not supported. Across all nine augmentations, the per-model Spearman correlations between symmetric envelope-spectrum fault-band fidelity and gap recovery are positive but non-significant:  $\rho = 0.4333$ ,  $p = 0.2440$  for 1D and  $\rho = 0.3167$ ,  $p = 0.4064$  for 2D (Figure 13). Because the negative control has extreme recovery values, we repeat the analysis without it; correlations fall to  $\rho = 0.1905$  ( $p = 0.6514$ ) and  $\rho = 0.0238$  ( $p = 0.9554$ ). An exploratory pooled value is not used inferentially because both models reuse the same nine fidelity measurements.

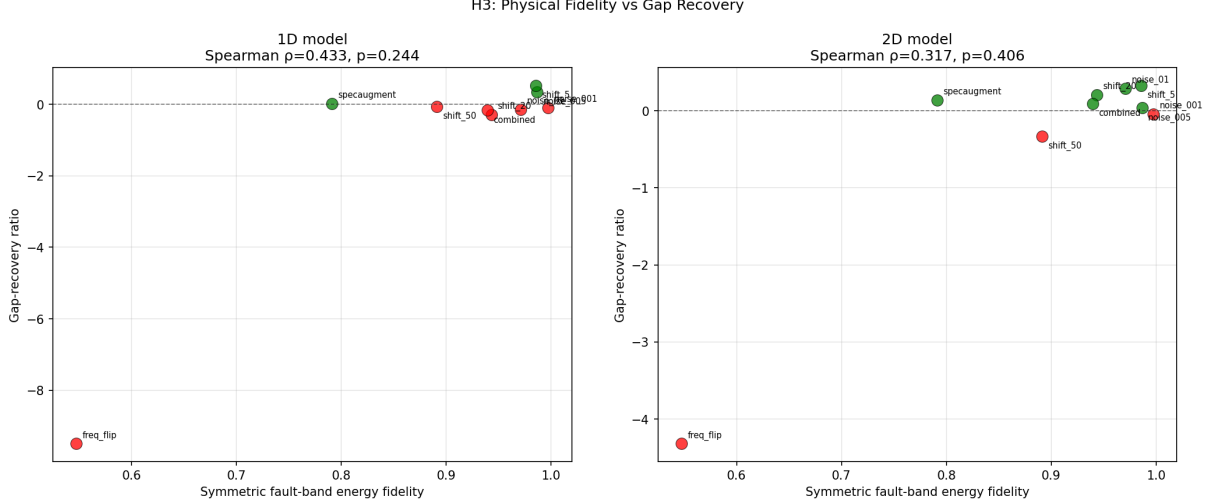


Figure 13: H3: Symmetric physical energy fidelity ( $F_{\text{energy}}$ ) vs. gap recovery ( $\Delta_{\text{rec}}$ ), reported separately by model.

**Exploratory contamination check.** A paired, single-seed auxiliary test evaluates selected recording-level models on common clean and noise-contaminated test realizations up to  $\sigma = 0.10$ . It shows little degradation over the tested range, but one seed cannot establish statistical robustness or rank augmentation strategies. The complete values are retained in `results/tables/contamination_robustness.json` and are not used to support H1–H3.

## 9 Ablation

### 9.1 Single vs. Combined Augmentation

Combined augmentation does not outperform the best single augmentation. Under recording-level splitting, combined reaches 0.9334 for 1D and 0.8960 for 2D, whereas the 5-sample shift reaches 0.9728 and 0.9229. This result does not show that stacking is generally harmful, but it provides no benefit for the tested noise-plus-shift combination.

### 9.2 Grouping Stress-Test Comparison

The fault-size strategy has the lowest observed accuracy (0.6508/0.4920), but it also trains on one of three diameter groups and only 14 recordings. Table 6 therefore identifies a demanding configuration rather than causal variable importance.

## 10 Discussion

**H1 (partial recovery)** is supported. The best tested augmentation, a 5-sample shift, recovers 52.55% of the 1D protocol gap and 32.63% of the 2D gap; neither closes the gap.

**H2 (protocol-gap dominance)** is supported operationally: the largest recovery ratio remains below one for both models. This experiment cannot assign a causal fraction of the gap to exact shared-sample leakage versus the broader unseen-recording distribution shift.

**H3 (physical fidelity)** is not supported. Neither per-model test is significant, with or without the negative control. This null result does not establish that physical fidelity is irrelevant; nine augmentation-level observations provide limited power and the metric captures only envelope-spectrum band energy.

**Seed-count sensitivity.** Using only the first three configured seeds would give recording baselines of 0.9961 (1D) and 0.9631 (2D). Including seeds four and five changes the five-seed

means to 0.9429 and 0.8856, shifts of 5.32 and 7.75 percentage points. Five seeds still do not guarantee stable population estimates, but the observed change shows that three were inadequate here.

**Fault-size stress test.** Its low 0.6508/0.4920 accuracy warrants follow-up with a matched-coverage design. The present comparison cannot separate fault-size shift from the fact that this strategy trains on fewer recordings and only one diameter group.

## 11 Limitations

1. **Single benchmark and artificial faults:** All experiments use CWRU’s seeded laboratory faults. High accuracy here does not establish field readiness or performance on naturally evolving damage.
2. **Condition dependence:** Only the drive-end 12 kHz accelerometer is used. Results may change with load, speed, sensor location, sampling rate, or segmentation; fan-end 48 kHz data would provide one sensitivity check.
3. **Leakage-prone comparator:** Random adjacent-window splitting intentionally permits source recordings and shared samples to cross partitions. Its result is a diagnostic comparator, not an estimate of unseen-recording performance.
4. **Fixed augmentation parameters:** Parameters chosen a priori; sensitivity not explored.
5. **Offline augmentation:** Each transform is sampled once per run and replaces, rather than expands, the training partition; online per-epoch resampling may behave differently.
6. **CNN architectures only:** Results may differ for Transformer or MLP architectures.
7. **50-epoch ceiling:** Eight of 200 runs reach their best validation accuracy in epochs 45–50; this best-epoch diagnostic does not prove formal convergence.
8. **Physical metric and sample size:** Symmetric band-energy fidelity covers one physical dimension, and H3 has only nine augmentation-level observations per model.
9. **Grouping comparability:** Group counts and training-recording counts differ, so grouping accuracy cannot be interpreted as causal variable importance.
10. **Reduced robustness scope:** The planned all-pairs cross-load experiment was not completed, and the contamination check uses one seed. Neither supports a general robustness claim.
11. **AI-assisted workflow:** AI assisted coding, analysis, drafting, and review. Artifact-level verification reduces but cannot eliminate transcription, framing, or citation-audit error; responsibility for the claims remains with the authors.

## 12 Conclusion

Matched-preprocessing evaluation on CWRU yields no-augmentation random-to-recording protocol gaps of 0.0569 (1D) and 0.1143 (2D). A 5-sample shift recovers 52.55% and 32.63%, while the destructive negative control collapses performance. H3 is not supported in either model or in the sensitivity analysis without the control. Fault-size grouping is the hardest observed stress test, but unequal coverage prevents a causal ranking. The seed diagnostic, exact split assignments, complete logs, matrices, and generated tables make the limits as well as the findings independently checkable.

## 13 Reproducibility Statement

All code, selected-file hashes, exact split manifests, experimental configurations, and results are included in the project repository. Random seeds (42, 123, 456, 789, 1024) are fixed across all experiments. The exact A10 package versions are recorded in the repository’s `environment`

directory, in `requirements-a10-lock.txt`. Set `CWRU_DATA_ROOT` to the directory containing the selected source files and, for GPU execution, set `CWRU_DEVICE=cuda`. Run:

```
python src/run_all.py
python src/verify_release.py
```

The first command regenerates the evidence, while the second checks run counts, configurations, matrices, manifests, generated tables, figures, and PDF freshness. Per-epoch logs for all 200 main runs are included. Raw CWRU files are not redistributed.

## 14 AI-Use Disclosure

The work used DeepSeek via OpenCode CLI for initial pair-programming, debugging, analysis, and drafting. OpenAI Codex subsequently performed an independent course-compliance and data-copy audit, identified protocol/evidence mismatches, implemented the documented repairs, orchestrated an isolated A10 rerun, and reconciled the release artifacts. The complete interaction summary and verification trail are recorded in `prompts/prompt_log.md`. Numerical statements were checked against generated JSON/CSV artifacts, dataset statements against the material passport and manifests, and citations against the retained bibliography/source records. Numerical results and figures come from code executed on the selected CWRU files; AI did not invent missing results. The authors retain responsibility for the final interpretation and disclosure.

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